DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY

CHE 450 **Homework Set 3**

Problem 1 (25%) You are interested in the production of C from A and B according to the following reaction:

$$A\ (\text{vapor}) + 2\ B\ (\text{gas}) \rightarrow 2\ C\ (\text{vapor}) + D\ (\text{vapor})$$

Assume that Species A and C condense at temperatures between 50-60°C, while Species B is a permanent gas and Species D only condenses at temperatures below 0°C. Species A is stored on the plant site as a liquid, which must be pumped and heated to high-temperature / high-pressure reaction conditions. Species B is stored as a compressed gas and must similar be adjusted to reaction conditions. Species B is harmful to the environment. The single pass conversion of Species A through the reactor is 95%. Feedstock / product pricing information is given in the table below:

Species	Price ($/tonne)	MW (g/mol)
A	1500	90
B	500	70
C	1200	100
D	400	30

(a, 5%) Sketch a process concept diagram for a proposed process utilizing this reaction.
(b, 10%) Sketch a block flow diagram for the process, including any recycle streams. Assume there is only one reactor in the system.
(c, 10%) Perform a due-diligence economic analysis of the process concept. Based on the information provided, do you recommend further investigation into the economic feasibility of this process? Why?

Problem 2 (18%) As we discussed in class, when performing initial design of a process there are three forms of recycle structure for unused reactants. For each of these cases we discussed in class, carefully explain under what conditions you would consider implementing each strategy:
(a, 6%) Separate and purify
(b, 6%) Recycle feed and product together (with purge stream)
(c, 6%) Recycle feed and product together (without purge stream)

Problem 3 (24%) For each of the following cases, state whether you would specify a batch or continuous process and provide justification (i.e. explain why):
(a, 8%) A facility producing 10W-40 oil for worldwide (>5000 tonne/yr) distribution
(b, 8%) A facility generating vaccines for worldwide (>5000 tonne/yr) distribution
(c, 8%) A small contract chemical company which produces various quantities of different specialty chemicals when clients place orders.

Problem 4 (24%) For each of the following scenarios, specify whether it would be appropriate to (a) pay for steam only, (b) pay for steam and boiler feed water, or (c) receive credit for steam produced. In addition, explain your rationale for each of your answers.

(a, 8%) Low pressure steam is fed through a pipe to remove oils clinging to the inner pipe wall. The steam and entrained oils exit the pipe to the atmosphere.
(b, 8%) Boiler feed water flows in the shell side of a shell-and-tube heat exchanger. The tubes of the heat exchanger act as packed bed reactors where a highly exothermic reaction takes place. The heat released by the exothermic reaction in the tubes is removed by generating medium pressure steam in the shell, which then exits the heat exchanger and is returned to the steam generator.
(c, 8%) Your facility uses high-pressure steam in the tube side of a shell-and-tube heat exchanger to heat a hydrocarbon mixture. The steam condenses in the tubes, exiting the heat exchanger as liquid water which is returned to the steam generator.

Problem 5 (9%) For the case in Problem 4 Part (b), would it be good practice to generate steam using cooling water and return this steam to the steam generator? Why or why not?